



Name: _____

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Learning Objectives

- Find indefinite integrals using the Power Rule and basic rules
- Evaluate definite integrals using the Fundamental Theorem of Calculus
- Apply u-substitution to evaluate composite integrals
- Interpret definite integrals in applied contexts

Always include $+C$ for indefinite integrals. For u-substitution, show your substitution clearly and convert the limits when evaluating definite integrals.

1. Find the indefinite integral.

$$\int (4x^3 - 6x + 5) dx$$

Answer: _____

2. Evaluate the definite integral.

$$\int_1^3 (2x^2 - x + 4) dx$$

Answer: _____

3. Find the indefinite integral of the exponential function.

$$\int \left(3e^x + \frac{2}{x} \right) dx$$

Answer: _____

4. Evaluate using u-substitution. Let $u = 2x + 3$.

$$\int (2x + 3)^5 dx$$

Answer: _____

5. Evaluate the definite integral.

$$\int_0^{\pi} \sin x dx$$

Answer: _____

6. Use u-substitution to evaluate.

$$\int_0^1 x e^{x^2} dx$$

Answer: _____



7. Evaluate the definite integral using the Fundamental Theorem of Calculus.

$$\int_1^4 \frac{3}{\sqrt{x}} dx$$

Answer: _____

8. Find the indefinite integral by rewriting the integrand.

$$\int \frac{x^3 - 2x}{x} dx$$

Answer: _____

9. The rate of change of a population is $P'(t) = 200 + 40t$ people/year. Find the total growth from $t = 0$ to $t = 5$.

$$\int_0^5 (200 + 40t) dt$$

Answer: _____

10. Evaluate using u-substitution.

$$\int \frac{4x}{x^2 + 1} dx$$

Answer: _____





Problem 4 and 6 use u-sub with definite integrals — remind students to convert limits. Problem 8 requires simplifying the integrand algebraically first. Problem 9 is an applied rate/accumulation problem.

Solutions

1. Find the indefinite integral.

$$\int (4x^3 - 6x + 5) dx$$

→ Apply the Power Rule: $\int x^n dx = x^{n+1}/(n+1) + C$.

→ $\int 4x^3 dx = x^4$. $\int -6x dx = -3x^2$. $\int 5 dx = 5x$.

→ Result: $x^4 - 3x^2 + 5x + C$.

Answer: $x^4 - 3x^2 + 5x + C$

2. Evaluate the definite integral.

$$\int_1^3 (2x^2 - x + 4) dx$$

→ Antiderivative: $2x^3/3 - x^2/2 + 4x$.

→ At $x = 3$: $18 - 9/2 + 12 = 30 - 9/2 = 51/2$.

→ At $x = 1$: $2/3 - 1/2 + 4 = 25/6$.

→ $51/2 - 25/6 = 153/6 - 25/6 = 128/6 = 64/3$.

Answer: $\left[\frac{2x^3}{3} - \frac{x^2}{2} + 4x \right]_1^3 = \frac{92}{3}$

3. Find the indefinite integral of the exponential function.

$$\int \left(3e^x + \frac{2}{x} \right) dx$$

→ $\int e^x dx = e^x$. $\int (1/x) dx = \ln|x|$.

→ Result: $3e^x + 2 \ln|x| + C$.

Answer: $3e^x + 2 \ln|x| + C$

4. Evaluate using u-substitution. Let $u = 2x + 3$.

$$\int (2x + 3)^5 dx$$

→ Let $u = 2x+3$, $du = 2dx \rightarrow dx = du/2$.

→ $\int u^5 \cdot (du/2) = (1/2) \cdot u^6/6 + C$.

→ $= (2x+3)^6/12 + C$.

Answer: $\frac{(2x + 3)^6}{12} + C$



5. Evaluate the definite integral.

$$\int_0^{\pi} \sin x \, dx$$

$$\rightarrow \int \sin x \, dx = -\cos x.$$

$$\rightarrow \text{At } x = \pi: -\cos \pi = -(-1) = 1.$$

$$\rightarrow \text{At } x = 0: -\cos 0 = -1.$$

$$\rightarrow \text{Result: } 1 - (-1) = 2.$$

Answer: $[-\cos x]_0^{\pi} = 2$

6. Use u-substitution to evaluate.

$$\int_0^1 x e^{x^2} \, dx$$

$$\rightarrow \text{Let } u = x^2, \, du = 2x \, dx \rightarrow x \, dx = du/2.$$

$$\rightarrow \text{Limits: } x=0 \rightarrow u=0; \, x=1 \rightarrow u=1.$$

$$\rightarrow = (1/2) \int_0^1 e^u \, du = (1/2)[e^u]_0^1.$$

$$\rightarrow = (1/2)(e - 1).$$

Answer: $\frac{1}{2}(e - 1)$

7. Evaluate the definite integral using the Fundamental Theorem of Calculus.

$$\int_1^4 \frac{3}{\sqrt{x}} \, dx$$

$$\rightarrow \text{Rewrite: } 3x^{-1/2}. \text{ Antiderivative: } 3 \cdot x^{1/2} / (1/2) = 6\sqrt{x}.$$

$$\rightarrow \text{At } x = 4: 6(2) = 12. \text{ At } x = 1: 6(1) = 6.$$

$$\rightarrow \text{Result: } 12 - 6 = 6.$$

Answer: $[6\sqrt{x}]_1^4 = 6$

8. Find the indefinite integral by rewriting the integrand.

$$\int \frac{x^3 - 2x}{x} \, dx$$

$$\rightarrow \text{Divide each term: } (x^3 - 2x)/x = x^2 - 2.$$

$$\rightarrow \int (x^2 - 2) \, dx = x^3/3 - 2x + C.$$

Answer: $\frac{x^3}{3} - 2x + C$

9. The rate of change of a population is $P'(t) = 200 + 40t$ people/year. Find the total growth from $t = 0$ to $t = 5$.

$$\int_0^5 (200 + 40t) \, dt$$

$$\rightarrow \text{Antiderivative: } 200t + 20t^2.$$

$$\rightarrow \text{At } t = 5: 1000 + 500 = 1500.$$

$$\rightarrow \text{At } t = 0: 0.$$

$$\rightarrow \text{Total growth} = 1500 \text{ people.}$$

Answer: $[200t + 20t^2]_0^5 = 1500$



10. Evaluate using u-substitution.

$$\int \frac{4x}{x^2 + 1} dx$$

→ Let $u = x^2 + 1$, $du = 2x dx \rightarrow 4x dx = 2 du$.

→ $\int (2/u) du = 2 \ln|u| + C$.

→ $= 2 \ln(x^2 + 1) + C$. ($x^2 + 1 > 0$, no absolute value needed)

Answer: $2\ln(x^2 + 1) + C$

